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Sem - 3rd

Course - Btech CSE (Cybersecurity)

OBJECT ORIENTED PROGRAMMING LAB ASSIGNMENT -1 (Topics – Basics of C++, Control Structures)

Programming Questions

1. Write a program (WAP) to display "Hello World" on console display. WAP to implement the following control characters:

- \n is for new line, or you can use endl –
cout<<endl<<"message";
- \t is for tab
- \a is an alarm sound
- \r is carriage return to go to the beginning of the current line.

Sol.

Code: #include <iostream>

using namespace std;

int main()

{

```
cout << "Hello World";  
cout << "\nHere I implemented newline";  
cout << endl << "Here I implemented endl" << endl;  
cout << "Here\tI\timplemented\ttab" << endl ;  
cout << "Here I implemented alarm \a \a \a";  
cout << endl << "There I implemented carriage return\rHere ";  
cout << endl ;//to flush the /r effecttttttt
```

return 0;

```
}
```

Output:

```
PS> & "C:\Users\meow\Documents\University\C++ OOPS\Ass1.exe"

Select Question from 1-13 (Enter 0 to EXIT) : 1
Hello World
Here I implemented newline
Here I implemented endl
Here I implemented tab
Here I implemented alarm
Here I implemented carriage return

Select Question from 1-13 (Enter 0 to EXIT) : █
```

2. Write a C++ program that will ask for a temperature in Celsius and display it in degree Fahrenheit.[$F=9C/5+32$]

Sol.

Code: #include <iostream>

using namespace std;

int main()

{

```
float C,F;
    cout << endl << "Enter Temp in C : ";
    cin >> C ;
    F=(9*C)/5+32;
    cout << "Temp in F : " << F;
```

return 0;

}

Output:

```
Select Question from 1-13 (Enter 0 to EXIT) : 2
```

```
Enter Temp in C : 36.67
```

```
Temp in F : 98.006
```

3. WAP to calculate gross salary of a person, where gross_salary = basic+da+ta+HRA and da is 46% of basic and ta is 8% of basic and HRA is 9% of basic pay.

Sol.

Code: #include <iostream>

using namespace std;

int main()

{

```
float base,gorss,da,ta,hra ;  
cout << endl << "Enter Basic Salary : ";  
cin >> base ;  
da = 0.46 * base ;  
ta = 0.08 * base ;  
hra = 0.09 * base ;  
gorss = base + da + ta + hra ;  
cout << "Gross Salary : " << gorss;
```

return 0;

}

Output:

```
Select Question from 1-13 (Enter 0 to EXIT) : 3
```

```
Enter Basic Salary : 100000
```

```
Gross Salary : 163000
```

4. A certain grade of steel is graded according to the following conditions: (i) Hardness must be greater than 50 (ii) Carbon content must be less than 0.7 (iii) Tensile strength must be greater than 5600 The grades are as follows: Grade is 10 if all three conditions are met Grade is 9 if all conditions (i) and (ii) are met Grade is 8 if all conditions (ii) and (iii) are met Grade is 7 if all conditions (i) and (iii) are met Grade is 6 if only one condition is met Grade is 5 if none of the conditions are met Write a program, which will require the user to give values of hardness, carbon content and tensile strength of the steel under consideration and output the grade of the steel. (Solve using nested if-else, ladder if and else if)

Sol.

(using nested if-else)

Code: #include <iostream>

using namespace std;

int main()

{

```
float h,t,c;
cout << endl <<"Enter Hardness : ";
cin >> h;
cout << endl <<"Enter Carbon Content : ";
cin >> c;
cout << endl <<"Enter Tensile Strength : ";
cin >> t;
if ( h > 50 )
{
    if ( c < 0.7)
    {
        if (t > 5600)
            cout << "\nSteel Quality : Grade 10";
        else
            cout << "\nSteel Quality : Grade 9";
    }
    else
    {
        if (t>5600)
            cout << "\nSteel Quality : Grade 7";
        else
            cout << "\nSteel Quality : Grade 6";
    }
}
```

```

    }
    else
    {
        if (c<0.7)
        {
            if (t>5600)
                cout << "\nSteel Quality : Grade 8";
            else
                cout << "\nSteel Quality : Grade 6";
        }
        else if (t>5600)
            cout << "\nSteel Quality : Grade 6";
        else
            cout << "\nSteel Quality : Grade 5";
    }
}

```

return 0;

}

Output:

```

Enter Hardness : 50

Enter Carbon Content : 0.2

Enter Tensile Strength : 40000

Steel Quality : Grade 8

```

(Using if-else ladder)

Code: #include <iostream>

using namespace std;

int main()

{

float h, c, t;

```
int a, g;
```

```
cout << "Enter hardness: ";
```

```
cin >> h;
```

```
cout << "Enter carbon content: ";
```

```
cin >> c;
```

```
cout << "Enter tensile strength: ";
```

```
cin >> t;
```

```
a = (h > 50) + (c < 0.7) + (t > 5600);
```

```
if (a == 3)
```

```
    g = 10;
```

```
else if (h > 50 && c < 0.7)
```

```
    g = 9;
```

```
else if (c < 0.7 && t > 5600)
```

```
    g = 8;
```

```
else if (h > 50 && t > 5600)
```

```
    g = 7;
```

```
else if (a == 1)
```

```
    g = 6;
```

```
else
```

```
    g = 5;
```

```
cout << "Grade = " << g;
```

```
return 0;
```

```
}
```

Output:

```
Enter hardness: 67
Enter carbon content: 4
Enter tensile strength: 6666
Grade = 7
```

(Using else-if conditions)

Code: #include <iostream>

using namespace std;

int main()

{

float h, c, t;

int g;

cout << "Enter hardness: ";

cin >> h;

cout << "Enter carbon content: ";

cin >> c;

cout << "Enter tensile strength: ";

cin >> t;

if (h > 50 && c < 0.7 && t > 5600)

g = 10;

else if (h > 50 && c < 0.7)

g = 9;

```

    else if (c < 0.7 && t > 5600)
    g = 8;
else if (h > 50 && t > 5600)
    g = 7;
else if ((h > 50) + (c < 0.7) + (t > 5600) == 1)
    g = 6;
else
    g = 5;

cout << "Grade = " << g;

return 0;
}

```

Output:

```

Enter hardness: 67
Enter carbon content: 4
Enter tensile strength: 6666
Grade = 7

```

5. A library charges a fine for every book return late. For first 5 days the fine is 50 paisa, for 6-10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled. WAP to accept no. of days the member is late to return the book and display the fine or appropriate message.

Sol.

Code:

```

#include <iostream>

using namespace std;

```

```
int main()
```

```
{
```

```
    unsigned int days;
    cout << endl << "Enter No. of Late Days : ";
    cin >> days ;
    if (days == 0)
        cout << "\nFine : Zero";
    else if (days <= 5)
        cout << "\nFine : 50 paisa";
    else if (days <= 10)
        cout << "\nFine : 1 rupee";
    else if (days < 30)
        cout << "\nFine : 5 rupees";
    else
        cout << "\nMEMBERSHIP CANCELLED ! \nFine : 5 rupees\n";
```

```
    return 0;
```

```
}
```

Output:

```
Select Question from 1-13 (Enter 0 to EXIT) : 5

Enter No. of Late Days : 31

MEMBERSHIP CANCELLED !
Fine : 5 rupees
```

6. WAP to demonstrate for, while, do-while (with all possible variations), like for loop can be demonstrated without giving initialization in for construct or without giving increment in for construct.

Sample: for (int i=0; i<10; i++)

i=0

```
for (; i<10; i++)
```

```
i=0
```

```
for (; i<10;)
```

```
i++
```

```
Code: #include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    cout << endl << "\nFor Loop\nCase I\n";
    for(int i=0;i<5;i++)
    {cout << endl << i ;}

    cout << endl << "\nFor Loop\nCase II\n";
    int i = 0;
    for(;i<5;i++)
    {cout << endl << i ;}

    cout << endl << "\nFor Loop\nCase III\n";
    int j;
    for(j=0;j<5;j++)
    {cout << endl << j ;}

    cout << endl << "\nFor Loop\nCase IV\n";
    int k = 0;
    for(;k<5;)
    {cout << endl << k ;k++;}

    cout << endl << "\nWhile Loop\nCase I\n";
    int l = 0;
    while(l<5)
    {cout << endl << l ;l++;}

    cout << endl << "\nDo-While Loop\nCase I\n";
```

```
int h = 0;
do
{cout << endl << h ;h++;}while(h<5);
```

return 0;

}

Output:

```
Select Question from 1-13 (Enter 0 to EXIT) : 6

For Loop
Case I

0
1
2
3
4

For Loop
Case II

0
1
2
3
4

For Loop
Case III

0
1
2
3
4

For Loop
Case IV

0
1
2
3
4

While Loop
Case I

0
1
2
3
4

Do-While Loop
Case I

0
1
2
3
4
Select Question from 1-13 (Enter 0 to EXIT) : █
```

7. Implement namespace in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.

Sol. Code: #include <iostream>

using namespace std;

```
namespace meow
{
    string var = "Yoooooooooooooooo Broooooooooooooo !";

    void q7()
    {
        cout << endl << "I'm a fn built inside a namespace !";
    }
}
```

int main()

{

```
meow::q7();
    cout << "\nI'm a variable inside namespace : " << meow::var;
```

return 0;

}

Output:

```
Select Question from 1-13 (Enter 0 to EXIT) : 7

I'm a fn built inside a namespace !
I'm a variable inside namespace : Yoooooooooooooooo Broooooooooooooo !
```

8. Create a structure in C++ containing the details of Students as details below and a main function to execute the structure.

Data Members(properties):

Name Roll No Degree Hostel CurrentCGPA

Member Function(behavior):

addDetails(); updateDetails(); updateCGPA(); updateResidenceInfo();
displaydetails();

Sol. Code:

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
struct Student
```

```
{
```

```
    string name;
```

```
    int roll;
```

```
    string degree;
```

```
    string hostel;
```

```
    float cgpa;
```

```
    void addDetails()
```

```
    {
```

```
        cout << "Enter name: ";
```

```
        cin >> name;
```

```
        cout << "Enter roll number: ";
```

```
        cin >> roll;
```

```
cout << "Enter degree: ";
```

```
cin >> degree;
```

```
cout << "Enter hostel: ";
```

```
cin >> hostel;
```

```
cout << "Enter CGPA: ";
```

```
cin >> cgpa;
```

```
}
```

```
void updateDetails()
```

```
{
```

```
    cout << "Enter new name: ";
```

```
    cin >> name;
```

```
    cout << "Enter new degree: ";
```

```
    cin >> degree;
```

```
}
```

```
void updateCGPA()
```

```
{
```

```
    cout << "Enter new CGPA: ";
```

```
    cin >> cgpa;
```

```
}
```

```
void updateResidenceInfo()
{
    cout << "Enter new hostel: ";
    cin >> hostel;
}
```

```
void displaydetails()
{
    cout << "\nStudent Details\n";
    cout << "Name: " << name << endl;
    cout << "Roll No: " << roll << endl;
    cout << "Degree: " << degree << endl;
    cout << "Hostel: " << hostel << endl;
    cout << "CGPA: " << cgpa << endl;
}
};
```

```
int main()
{
    Student s;

    s.addDetails();
    s.displaydetails();

    s.updateCGPA();
    s.updateResidenceInfo();
}
```

```

        cout << "\nUpdated Details:\n";

        s.displaydetails();

        return 0;
    }

```

Output:

```

Degree: btech
Hostel: brs
CGPA: 9.8
Enter new CGPA: 8.5
Enter new hostel: bms

Updated Details:

Student Details
Name: Ravi
Roll No: 67
Degree: btech
Hostel: bms
CGPA: 8.5

```

9. Differentiate between private and public access/scope. Perform the question no. 5 with structure having the data member in private scope. Students should be able to a) differentiate between structure in C vs structure in C++

Sol. Code:

```

#include <iostream>

using namespace std;

struct Library
{

```

private:

int d;

float f;

public:

void get()

{

cout << "Enter late days: ";

cin >> d;

}

void calculate()

{

if (d > 30)

{

cout << "Membership cancelled";

}

else if (d <= 5)

{

f = d * 0.50;

cout << "Fine = Rs. " << f;

}

else if (d <= 10)

{

f = d * 1;

cout << "Fine = Rs. " << f;

```

    }
    else
    {
        f = d * 5;
        cout << "Fine = Rs. " << f;
    }
}
};

```

```

int main()
{
    Library l;

    l.get();
    l.calculate();

    return 0;
}

```

Output:

```

Enter late days: 8
Fine = Rs. 8

```

10. Create a code snippet that illustrates the following:

- a. Calling of private member functions inside public member function**
- b. Access private member functions inside public member function**

Sol. Code: #include <iostream>

using namespace std;

```
class Demo
{
private:
    void show()
    {
        cout << "Private function called";
    }

public:
    void display()
    {
        show();
    }
};

int main()
{
    Demo d;

    d.display();

    return 0;
}
```

Output:

```
Private function called
```

11. An election is contested by five candidates. The candidates are numbered 1 to 5 and the voting is done by marking the candidate number on the ballot paper. Write a program to read the ballots and count the votes cast for each candidate using an array variable count. In case, a number read is outside the range 1 to 5, the ballot should be considered as a 'spoilt ballot', and the program should also count the number of spoilt ballots

Sol.

Code:

```
#include <iostream>

using namespace std;
```

```
int main()
{
    int n, v;
    int count[5] = {0, 0, 0, 0, 0};
    int spoilt = 0;

    cout << "Enter number of voters: ";
    cin >> n;

    for (int i = 0; i < n; i++)
    {
        cout << "Enter vote: ";
        cin >> v;

        if (v >= 1 && v <= 5)
        {
            count[v - 1]++;
        }
    }
}
```

```
    }  
    else  
    {  
        spoilt++;  
    }  
}
```

```
cout << "\nVotes of candidates:\n";
```

```
for (int i = 0; i < 5; i++)  
{  
    cout << "Candidate " << i + 1 << ": " << count[i] << endl;  
}
```

```
cout << "Spoilt ballots: " << spoilt;
```

```
    return 0;  
}
```

Output:

```

Enter vote: 3
Enter vote: 5
Enter vote: 6
Enter vote: 1
Enter vote: 5
Enter vote: 9

Votes of candidates:
Candidate 1: 2
Candidate 2: 1
Candidate 3: 1
Candidate 4: 0
Candidate 5: 2
Spoilt ballots: 2

```

12. Write programs to evaluate the following functions to 0.0001% accuracy.

$$(a) \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$(b) \text{SUM} = 1 + (1/2)^2 + (1/3)^3 + (1/4)^4 + \dots$$

$$(c) \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

Sol. Code: (a)

```
#include <iostream>
```

```
#include <cmath>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    double x, t, s = 0;
```

```
    int i;
```

```

cout << "Enter x in radians: ";
cin >> x;

for (i = 1; ; i += 2)
{
    t = pow(x, i) / tgamma(i + 1);

    if (i % 4 == 1)
        s = s + t;
    else
        s = s - t;

    if (fabs(t) < 0.000001)
        break;
}

cout << "sin x = " << s;

return 0;
}

```

Output:

```

Enter x in radians: 12
sin x = -0.536573

```

(b)

```

#include <iostream>

#include <cmath>

using namespace std;

```

```
int main()
{
    double s = 0, t;
    int i = 1;

    while (true)
    {
        t = pow(1.0 / i, i);
        s = s + t;

        if (t < 0.000001)
            break;

        i++;
    }

    cout << "SUM = " << s;

    return 0;
}
```

Output:

```
SUM = 1.29129
```

(c)

```
#include <iostream>
```

```
#include <cmath>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    double x, t, s = 0;
```

```
    int i;
```

```
    cout << "Enter x in radians: ";
```

```
    cin >> x;
```

```
    for (i = 0; ; i += 2)
```

```
    {
```

```
        t = pow(x, i) / tgamma(i + 1);
```

```
        if (i % 4 == 0)
```

```
            s = s + t;
```

```
        else
```

```
            s = s - t;
```

```
        if (fabs(t) < 0.000001)
```

```
            break;
```

```
    }
```

```
    cout << "cos x = " << s;
```

```
    return 0;
```

```
}
```

Output:

```
Enter x in radians: 45
cos x = 107.242
```

13. An electricity board charges the following rates to domestic users to discourage large consumption of energy:

For the first 100 units	60P per unit
For next 200 units	80P per unit
Beyond 300 units	90P per unit

All users are charged a minimum of Rs. 50.00. If the total amount is more than Rs. 300.00 then an additional surcharge of 15% is added. Write a program to read the names of users and number of units consumed and print out the charges with names

Sol.

Code:

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    string name;
```

```
    int u;
```

```
    float bill;
```

```
cout << "Enter name: ";
```

```
cin >> name;
```

```
cout << "Enter units consumed: ";
```

```
cin >> u;
```

```
bill = 50;
```

```
if (u <= 100)
```

```
{
```

```
    bill = bill + u * 0.60;
```

```
}
```

```
else if (u <= 300)
```

```
{
```

```
    bill = bill + 100 * 0.60;
```

```
    bill = bill + (u - 100) * 0.80;
```

```
}
```

```
else
```

```
{
```

```
    bill = bill + 100 * 0.60;
```

```
    bill = bill + 200 * 0.80;
```

```
    bill = bill + (u - 300) * 0.90;
```

```
}
```

```
if (bill > 300)
```

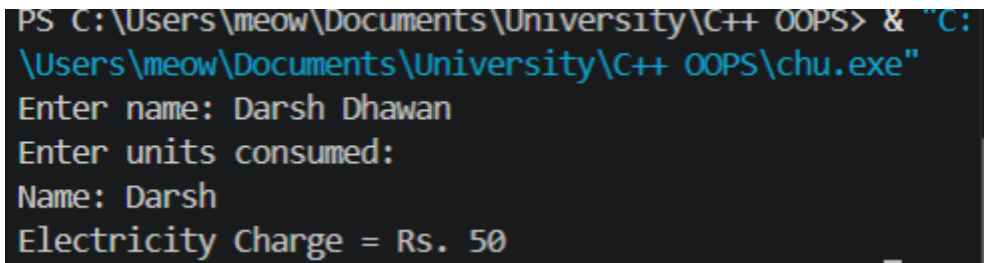
```
{
```

```
        bill = bill + bill * 0.15;
    }

    cout << "\nName: " << name;
    cout << "\nElectricity Charge = Rs. " << bill;

    return 0;
}
```

Output:



```
PS C:\Users\meow\Documents\University\C++ OOPS> & "C:\Users\meow\Documents\University\C++ OOPS\chu.exe"
Enter name: Darsh Dhawan
Enter units consumed:
Name: Darsh
Electricity Charge = Rs. 50
```